Homework assignment for 02YMECH
Chapter: Dynamics

Solutions

1. **Child throwing a package from a moving boat.**

   We have three masses: child m_c, boat m_b, and package m_p. The water is assumed frictionless in the horizontal directions during the throw. We take the water as an inertial lab frame.

   Let $\vec{v}_0$ be the initial velocity of the entire system (boat + child + package) before the throw. After the throw, the package has velocity $\vec{v}_p$ (given in the lab frame), and the boat+child system moves with velocity $\vec{v}_b$. We are asked to solve for $\vec{v}_b$ immediately after the throw.

   Total mass before the throw:

   $$M_{\text{tot}} = m_b + m_c + m_p.$$

   Because there are no external horizontal forces, linear momentum is conserved in the horizontal direction(s). (Vertically, gravity and normal force act, so we do not impose p_y conservation unless we assume no net external vertical impulse. We will keep both x and y components in the final answer, but note that strictly speaking only horizontal momentum is conserved. The standard idealization here assumes the throw is brief and any vertical impulse from the water on the boat averages to zero over the instant, so we treat total momentum as conserved as a vector in 2D.)

   Initial momentum:

   $$\vec{p}_i = M_{\text{tot}}\,\vec{v}_0.$$

   Final momentum:

   $$\vec{p}_f = (m_b + m_c)\,\vec{v}_b + m_p\,\vec{v}_p.$$

   Conservation of momentum:

   $$M_{\text{tot}}\,\vec{v}_0 = (m_b + m_c)\,\vec{v}_b + m_p\,\vec{v}_p.$$

   Solve for $\vec{v}_b$:

   $$\vec{v}_b = \frac{M_{\text{tot}}\,\vec{v}_0 - m_p\,\vec{v}_p}{m_b + m_c} = \frac{(m_b + m_c + m_p)\vec{v}_0 - m_p\vec{v}_p}{m_b + m_c}.$$

   Writing components explicitly, if

   $$\vec{v}_0 = (v_{0x}, v_{0y}), \qquad \vec{v}_p = (v_{px}, v_{py}), \qquad \vec{v}_b = (v_{bx}, v_{by}),$$

   then

   $$v_{bx} = \frac{(m_b + m_c + m_p)v_{0x} - m_p v_{px}}{m_b + m_c}, \qquad v_{by} = \frac{(m_b + m_c + m_p)v_{0y} - m_p v_{py}}{m_b + m_c}.$$

   If the package is thrown at speed $|\vec{v}_p|$ and angle θ above the horizontal, then

   $$v_{px} = |\vec{v}_p|\cos\theta, \qquad v_{py} = |\vec{v}_p|\sin\theta.$$

   Physically: the boat recoils so that the center of mass of (boat+child+package) keeps the same velocity that it had just before the throw.

2. **1D motion with position-dependent force.**

A particle of mass m moves along x under force

$$F(x) = -\frac{kx}{(x^2+a^2)^{3/2}}, \quad k > 0,\ a > 0.$$

Initially, at $x = x_0 > 0$, the particle is at rest.

(a) *Work done by the force from x_0 to x.*

The work done by a (possibly non-constant) force in 1D is

$$W(x_0 \to x) = \int_{x_0}^{x} F(u)\,du = \int_{x_0}^{x} \left[-\frac{ku}{(u^2+a^2)^{3/2}}\right] du.$$

Compute the integral. Let

$$I = \int \frac{u}{(u^2+a^2)^{3/2}}\,du.$$

Substitute $w = u^2 + a^2$, so $dw = 2u\,du$, i.e. $u\,du = \frac{1}{2}dw$. Then

$$I = \int \frac{1}{2} w^{-3/2} dw = \frac{1}{2}\int w^{-3/2} dw = \frac{1}{2}\left(\frac{w^{-1/2}}{-1/2}\right) = -w^{-1/2}.$$

So

$$I = -\frac{1}{\sqrt{u^2+a^2}} + C.$$

Therefore,

$$\int_{x_0}^{x} \left[-\frac{ku}{(u^2+a^2)^{3/2}}\right] du = -k\int_{x_0}^{x} \frac{u}{(u^2+a^2)^{3/2}} du = -k\left[-\frac{1}{\sqrt{u^2+a^2}}\right]_{u=x_0}^{u=x}.$$

This gives

$$W(x_0 \to x) = k\left[\frac{1}{\sqrt{u^2+a^2}}\right]_{x_0}^{x} = k\left(\frac{1}{\sqrt{x^2+a^2}} - \frac{1}{\sqrt{x_0^2+a^2}}\right).$$

Sign check: if $x < x_0$ (the particle moves toward the origin), then typically the work is positive because the force points toward $x = 0$ for $x > 0$.

(b) *Velocity as a function of x (work-energy theorem).*

The work-energy theorem in 1D:

$$W(x_0 \to x) = \Delta K = \frac{1}{2}mv^2(x) - \frac{1}{2}mv^2(x_0).$$

We are told the particle starts from rest, so $v(x_0) = 0$. Then

$$\frac{1}{2}mv^2(x) = W(x_0 \to x) = k\left(\frac{1}{\sqrt{x^2+a^2}} - \frac{1}{\sqrt{x_0^2+a^2}}\right).$$

Solve for $v(x)$:

$$v^2(x) = \frac{2k}{m}\left(\frac{1}{\sqrt{x^2+a^2}} - \frac{1}{\sqrt{x_0^2+a^2}}\right),$$

$$v(x) = \sqrt{\frac{2k}{m}\left(\frac{1}{\sqrt{x^2+a^2}} - \frac{1}{\sqrt{x_0^2+a^2}}\right)}.$$

Note: the expression in parentheses must be nonnegative for real v. Since the particle starts at $x_0 > 0$ and is attracted toward $x = 0$, x will decrease from x_0 toward 0, and $1/\sqrt{x^2+a^2}$ increases as $|x|$ decreases.

(c) *Maximum speed.*

The speed increases as the particle moves to where the kinetic energy is largest. From the expression above, $v^2(x)$ is largest when

$$\frac{1}{\sqrt{x^2+a^2}} \quad \text{is largest.}$$

That happens at the smallest possible $\sqrt{x^2+a^2}$, i.e. at $x = 0$ (symmetric force will then decelerate it once it passes through and climbs "up" on the other side).

Thus the maximum speed $v_{\max}$ occurs at $x = 0$:

$$v_{\max}^2 = \frac{2k}{m}\left(\frac{1}{\sqrt{0+a^2}} - \frac{1}{\sqrt{x_0^2+a^2}}\right) = \frac{2k}{m}\left(\frac{1}{a} - \frac{1}{\sqrt{x_0^2+a^2}}\right).$$

Therefore

$$v_{\max} = \sqrt{\frac{2k}{m}\left(\frac{1}{a} - \frac{1}{\sqrt{x_0^2+a^2}}\right)}.$$

Interpretation: the force pulls the particle toward the origin like a softened $1/r^2$-type attraction; the speed peaks at the center.

3. **Inelastic capture on a rotating disk.**

A disk of mass M and radius R, initially at rest, can rotate freely about its symmetry axis (perpendicular to its plane). A particle of mass m moves tangentially with speed v_0 and sticks to the rim. The disk's moment of inertia about its axis is

$$I_{\text{disk}} = \frac{1}{2}MR^2.$$

We assume no external torque about the axis during the sticking, so angular momentum about the axis is conserved.

Before collision:

- Disk is at rest $\Rightarrow$ zero angular momentum.
- Particle of mass m moves tangentially at radius R, so its angular momentum about the disk center is

  $$L_i = mv_0R.$$

  (Sign chosen so that positive corresponds to the direction of rotation induced by the incoming particle.)

After collision:

- The particle sticks at the rim, becoming part of the rotating system.
- The final system is: disk + point mass m at radius R, both rotating with common angular velocity ω.

The final moment of inertia is

$$I_f = I_{\text{disk}} + I_{\text{particle}} = \frac{1}{2}MR^2 + mR^2 = R^2\left(\frac{1}{2}M + m\right).$$

(a) *Final angular velocity ω.*
Angular momentum conservation:

$$L_i = L_f \quad \Rightarrow \quad mv_0R = I_f\,\omega = R^2\left(\frac{1}{2}M + m\right)\omega.$$

Solve for ω:

$$\omega = \frac{mv_0R}{R^2\left(\frac{1}{2}M + m\right)} = \frac{mv_0}{R\left(\frac{1}{2}M + m\right)}.$$

(b) *Rotational kinetic energy after the collision.*
The final rotational kinetic energy is

$$T_{\text{rot}} = \frac{1}{2}I_f\omega^2 = \frac{1}{2}\left[R^2\left(\frac{1}{2}M + m\right)\right]\left[\frac{mv_0}{R\left(\frac{1}{2}M + m\right)}\right]^2.$$

Simplify:

$$T_{\text{rot}} = \frac{1}{2}R^2\left(\frac{1}{2}M + m\right)\frac{m^2v_0^2}{R^2\left(\frac{1}{2}M + m\right)^2} = \frac{1}{2}\frac{m^2v_0^2}{\left(\frac{1}{2}M + m\right)}.$$

So

$$T_{\text{rot}} = \frac{1}{2}\frac{m^2v_0^2}{\left(\frac{1}{2}M + m\right)}.$$

As expected for a perfectly inelastic capture, mechanical energy is not conserved: the final kinetic energy is less than the initial kinetic energy $\frac{1}{2}mv_0^2$, with the difference converted into internal energy/heat/deformation.

4. **Motion in a quartic potential.**

The potential energy is

$$U(x) = U_0\left(\frac{x^4}{a^4} - 2\frac{x^2}{a^2}\right) = U_0\left(\frac{x^4}{a^4} - \frac{2x^2}{a^2}\right), \quad U_0 > 0.$$

(a) *Force as a function of x.*
For 1D conservative motion,

$$F(x) = -\frac{\mathrm{d}U}{\mathrm{d}x}.$$

First compute $\frac{\mathrm{d}U}{\mathrm{d}x}$:

$$\frac{\mathrm{d}U}{\mathrm{d}x} = U_0\left(\frac{\mathrm{d}}{\mathrm{d}x}\frac{x^4}{a^4} - 2\frac{\mathrm{d}}{\mathrm{d}x}\frac{x^2}{a^2}\right) = U_0\left(\frac{4x^3}{a^4} - \frac{4x}{a^2}\right).$$

Therefore

$$F(x) = -\frac{\mathrm{d}U}{\mathrm{d}x} = -U_0\left(\frac{4x^3}{a^4} - \frac{4x}{a^2}\right) = -\frac{4U_0}{a^4}x^3 + \frac{4U_0}{a^2}x.$$

Factor if desired:

$$F(x) = \frac{4U_0}{a^4}\,x\left(a^2 - x^2\right).$$

(b) *Equilibrium points and their stability.*

Equilibrium occurs where $F(x) = 0$:

$$\frac{4U_0}{a^4}\, x\left(a^2 - x^2\right) = 0 \quad \Longrightarrow \quad x = 0 \quad \text{or} \quad a^2 - x^2 = 0.$$

So the equilibrium positions are

$$x = 0, \qquad x = +a, \qquad x = -a.$$

Stability can be determined from the curvature of $U(x)$: stable equilibrium at local minima of U, unstable at local maxima.

We inspect $U(x)$:

$$U(x) = U_0\left(\frac{x^4}{a^4} - \frac{2x^2}{a^2}\right).$$

Evaluate at the three critical points:

$$U(0) = 0, \qquad U(\pm a) = U_0\left(\frac{a^4}{a^4} - 2\frac{a^2}{a^2}\right) = U_0(1-2) = -U_0.$$

Since $U(\pm a) = -U_0 < 0 = U(0)$, $x = \pm a$ are lower than $x = 0$. So $x = \pm a$ are minima (stable equilibria), and $x = 0$ is a maximum (unstable equilibrium).

More formally, one could check $\frac{\mathrm{d}^2U}{\mathrm{d}x^2}$ at those points; $\frac{\mathrm{d}^2U}{\mathrm{d}x^2} > 0$ implies stable.

(c) *Speed at $x = a$ if released from rest at $x = 0$.*

Mechanical energy is conserved:

$$E = K + U = \text{constant}.$$

Initially ($x = 0$): released from rest $\Rightarrow v = 0$,

$$E = K_i + U_i = 0 + U(0) = U(0) = 0.$$

At $x = a$:

$$E = K_f + U(a) = \frac{1}{2}mv^2(a) + U(a).$$

Conservation $E_i = E_f$:

$$0 = \frac{1}{2}mv^2(a) + U(a).$$

We already have

$$U(a) = -U_0.$$

So

$$0 = \frac{1}{2}mv^2(a) - U_0 \quad \Longrightarrow \quad \frac{1}{2}mv^2(a) = U_0.$$

Solve for $v(a)$:

$$v(a) = \sqrt{\frac{2U_0}{m}}.$$

Note: the particle moves from an unstable equilibrium at $x = 0$ down to a stable equilibrium at $x = a$, losing potential energy (from 0 to $-U_0$) and gaining kinetic energy.

(d) *Work done by the force from $x = 0$ to $x = a$.*

The work done by a conservative force when moving from $x = 0$ to $x = a$ equals the negative change in potential energy:

$$W(0 \to a) = U(0) - U(a).$$

We have

$$U(0) = 0, \qquad U(a) = -U_0.$$

Therefore

$$W(0 \to a) = 0 - (-U_0) = U_0.$$

This agrees with the kinetic energy gain:

$$\Delta K = \frac{1}{2}mv^2(a) - 0 = U_0,$$

consistent with $v(a) = \sqrt{2U_0/m}$ found above.

5. **Explosion and conservation of momentum.**

Total mass before breakup:

$$M = 13 \text{ kg}.$$

Masses after breakup:

$$m_1 = 3 \text{ kg}, \quad m_2 = 4 \text{ kg}, \quad m_3 = 6 \text{ kg},$$

with speeds immediately after breakup:

$$\mathbf{v}_1 = (4 \text{ m/s})\,\hat{\imath} \quad \text{(east)}, \qquad \mathbf{v}_2 = (3 \text{ m/s})\,\hat{\jmath} \quad \text{(north)}.$$

Before breakup, the body was at rest, so initial total momentum was zero:

$$\mathbf{p}_{\text{total, initial}} = \mathbf{0}.$$

Momentum is conserved, so the vector sum of the fragment momenta after breakup must still be zero:

$$m_1\mathbf{v}_1 + m_2\mathbf{v}_2 + m_3\mathbf{v}_3 = \mathbf{0}.$$

Solve for $\mathbf{v}_3$:

$$m_3\mathbf{v}_3 = -\left(m_1\mathbf{v}_1 + m_2\mathbf{v}_2\right).$$

First compute the known momenta:

$$m_1\mathbf{v}_1 = (3 \text{ kg})(4 \text{ m/s})\,\hat{\imath} = 12\,\hat{\imath} \text{ kg} \cdot \text{m/s},$$

$$m_2\mathbf{v}_2 = (4 \text{ kg})(3 \text{ m/s})\,\hat{\jmath} = 12\,\hat{\jmath} \text{ kg} \cdot \text{m/s}.$$

Thus

$$m_1\mathbf{v}_1 + m_2\mathbf{v}_2 = 12\,\hat{\imath} + 12\,\hat{\jmath} \quad (\text{kg} \cdot \text{m/s}).$$

So

$$m_3\mathbf{v}_3 = -(12\,\hat{\imath} + 12\,\hat{\jmath}) = (-12)\,\hat{\imath} + (-12)\,\hat{\jmath}.$$

Since $m_3 = 6$ kg,

$$\mathbf{v}_3 = \frac{1}{6}(-12\,\hat{\imath} - 12\,\hat{\jmath}) = (-2)\,\hat{\imath} + (-2)\,\hat{\jmath} \quad \text{m/s}.$$

So the third fragment moves with velocity

$$\mathbf{v}_3 = (-2, -2) \text{ m/s}.$$

Speed:

$$v_3 = \sqrt{(-2)^2 + (-2)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \approx 2.83 \text{ m/s}.$$

Direction: both components are negative, so it moves toward the southwest, at 45° below the $-x$ axis (i.e. equally south and west).

We can express the direction angle θ measured from due west toward south:

$$\tan\theta = \frac{2}{2} = 1 \quad \Rightarrow \quad \theta = 45^\circ.$$

Thus m_3 moves to the southwest at $2\sqrt{2}$ m/s.

6. **Block on horizontal surface with time-dependent force.**

Given:

$$m = 4 \text{ kg}, \qquad \mu_s = 0.5, \qquad \mu_k = 0.3, \qquad g = 10 \text{ m/s}^2.$$

Applied horizontal force:

$$F(t) = 2t \quad \text{N}.$$

Normal force:

$$N = mg = (4)(10) = 40 \text{ N}.$$

Maximum static friction magnitude:

$$f_s^{\max} = \mu_s N = 0.5 \times 40 = 20 \text{ N}.$$

Kinetic friction magnitude (once sliding):

$$f_k = \mu_k N = 0.3 \times 40 = 12 \text{ N}.$$

(a) *When does the block start to move?*

While the block is at rest, friction adjusts to match the applied force $F(t)$ up to a maximum of 20 N. The block will start moving when $F(t)$ exceeds $f_s^{\max}$.

We solve

$$F(t) = 20 \text{ N} \quad \Rightarrow \quad 2t = 20 \quad \Rightarrow \quad t = 10 \text{ s}.$$

Interpretation: for $0 \le t < 10$ s, static friction is enough to hold it in place, so velocity is zero. At $t = 10$ s, motion begins.

(b) *Acceleration immediately after it starts moving.*

Just after $t = 10$ s, the block is in motion, so kinetic friction applies with constant magnitude $f_k = 12$ N opposing the motion.

The applied force at $t = 10$ s is

$$F(10) = 2(10) = 20 \text{ N}.$$

Net horizontal force just after motion begins:

$$F_{\text{net}} = F - f_k = 20 - 12 = 8 \text{ N}.$$

Newton's second law:

$$a = \frac{F_{\text{net}}}{m} = \frac{8}{4} = 2 \text{ m/s}^2.$$

So the block's initial acceleration (once it breaks free) is 2 m/s^2.

(c) *Velocity 5 seconds after it starts moving.*

After $t = 10$ s, the block continues to be pushed by $F(t) = 2t$. For $t > 10$ s, it is sliding, so friction is kinetic and constant at 12 N.

For $t > 10$ s:

$$F_{\text{net}}(t) = F(t) - f_k = 2t - 12.$$

Then the acceleration for $t > 10$ is

$$a(t) = \frac{F_{\text{net}}(t)}{m} = \frac{2t - 12}{4} = \frac{1}{2}t - 3 \quad (\text{m/s}^2).$$

Integrate from 10 to 15 (the velocity at $t = 10$ is 0):

$$\begin{aligned} v(15) - v(10) &= \int_{10}^{15} \left(\frac{1}{2}t - 3\right) dt \\ &= \left[\frac{1}{4}t^2 - 3t\right]_{10}^{15} \\ &= \frac{65}{4} \text{ m/s}. \end{aligned}$$